

Hardware Architecture

Digital Circuits

IDENTIFICATION

CODE : IF-3-AC
ECTS : 2.0

HOURS

Lectures : 9.0 h
Seminars : 4.0 h
Laboratory : 12.0 h
Project : 0.0 h
Teacher-student
contact : 25.0 h
Personal work : 20.0 h
Total : 45.0 h

ASSESSMENT METHOD

Written exam (1h30). Progress
evaluation by Moodle QCMs.

TEACHING AIDS

Lecture notes for the course, work
sheets for classwork and labs [in
French]

TEACHING LANGUAGE

French

CONTACT

M. BESLON Guillaume
guillaume.beslon@insa-lyon.fr

AIMS

The goal if this course is to gain theoretical and practical understanding of the principles and mechanisms that govern digital circuits, from basic logic gates up to a simple microprocessor.

Expected skills:

- coding and decoding information in binary
- building combinatorial circuits from boolean functions
- modeling sequential behavior as finite-state automata
- building a sequential circuit from a finite-state automaton
- evaluating the cost and performance of digital circuits

CONTENT

- coding: from information to bits
- computing: from boolean functions to combinatorial circuits
- storing: from latches to addressable memory
- controlling: from behavioral specification to sequential circuits (finite state automata)
- building complex circuits: separation of control and data
- Von Neuman model
- initiation to CAD and simulation tools for circuits

BIBLIOGRAPHY

- D. Patterson, J. Hennessy. Computer Organization and Design, The Hardware/Software Interface. Morgan Kaufmann.
- P. Amblard, JC. Fernandez, F. Lagnier, F. Maraninchi, P. Sicard, P. Waille. Architectures Logicielles et Matérielles. Dunod. [in french]